

The Keske Method

A Geometric Framework for Antibiotic Selection in Pediatric MRSA Acute Hematogenous Osteomyelitis

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Named for Steven Keske — 6 years, 5 antibiotics, 4 surgeries, still fighting.

The Clinical Gap

Acute hematogenous osteomyelitis (AHO) caused by community-associated MRSA (USA300 pulstotype) remains one of the most clinically treacherous pediatric infections. The United States sees approximately 70,000 pediatric musculoskeletal infections annually, with MRSA-AHO driving the most severe complications: septic shock, toxic shock syndrome (TSS), septic pulmonary emboli, pathologic fractures, and chronic relapsing disease requiring multi-year antibiotic courses.

The literature documents a reproducible failure mode: patients cycle through five or more antibiotic regimens over months to years, each serologically adequate, each clinically insufficient. Treatment guidelines remain empirical, and chronic biofilm-associated MRSA-AHO is widely regarded as refractory to standard antimicrobial therapy.

The Keske Method proposes that this failure is not primarily microbiological — it is geometric. The drugs exist. The standard pharmacokinetic models used to deploy them were built for blood, not bone. The math to deploy them correctly in deep-tissue sanctuaries is not part of routine clinical practice.

Governing Equation

The central question in antibiotic selection is: *will this drug reach a concentration high enough at the site of infection to kill the bacteria there?* The Keske Method formalizes that question as a single score, C_{bone} , computed from every known barrier between the IV bag and the bacteria inside bone. Higher C means fewer barriers, better penetration, higher probability of cure. The working cure threshold $C_{\text{bone}} \geq 5$ corresponds to achieving a bone concentration at or above the published MBEC for the dominant pathogen — the concentration required to eradicate biofilm, not merely suppress planktonic growth (Zimmerli et al., NEJM 2004; Stewart, Microbiol Mol Biol Rev 2015). This threshold is a modeling anchor derived from published data; it is not itself prospectively validated (see §7).

The Keske Method derives from the MIRADOR framework (Manifold-Informed Rational Architecture for Drug-Organism Response). The governing equation is:

$$C = \frac{\tau}{K}$$

τ (pharmacophoric potential) captures the drug's intrinsic activity: its binding affinity to the target (PBP2a for MRSA), aqueous solubility, oral or IV bioavailability, and protein binding. Clinically, τ is proportional to the drug's theoretical $\text{AUC}_{0-24}/\text{MIC}$ under ideal distribution conditions — the ceiling of what the drug can achieve if all barriers are removed.

K (total curvature) is the sum of all impedance factors that reduce the drug's effective concentration at the infection site. Clinically, K is what separates adequate serum levels from adequate bone levels. A drug with $K = 1$ loses 50% of its activity to barriers; a drug with $K = 9$ loses 90%.

For standard bacteremia, $K = K_{\text{ADMET}}$ (absorption, distribution, metabolism, excretion, toxicity). For pediatric AHO, three additional terms dominate:

$$K_{\text{bone}} = K_{\text{ADMET}} + K_{\text{penetration}} + K_{\text{biofilm}} + K_{\text{reservoir}}$$

Each term is clinically grounded, computationally specified, and sourced from published pharmacokinetic and microbiological literature. Every number in this brief is produced by the open-source MIRADOR engine and is reproducible from the inputs shown (§6).

The Four Curvature Layers

Layer 1: Pediatric Patient Manifold

Children are not small adults. The Keske Method replaces adult pharmacokinetic assumptions with pediatric-specific models:

$$\text{GFR}_{\text{ped}} = \frac{0.413 \times h_{\text{cm}}}{S_{\text{cr}}} \quad (\text{Schwartz equation}) \quad (1)$$

$$\text{CL}_{\text{ped}} = \text{CL}_{\text{adult}} \cdot \left(\frac{W}{70}\right)^{0.75} \quad (\text{allometric scaling}) \quad (2)$$

$$V_d = V_{d,\text{adult}} \cdot \left(\frac{W}{70}\right)^{1.0} \cdot [1 + 0.003 \cdot \max(\text{CRP} - 100, 0)] \quad (3)$$

The CRP-dependent V_d expansion (20–40% at $\text{CRP} > 100$ mg/L) reflects capillary leak in systemic inflammation (Graziani et al., JAC 1988).

Layer 2: Bone Penetration Curvature

The critical barrier that standard dosing models ignore is the serum-to-bone partitioning ratio R_{bone} . Published bone:serum ratios are drug-specific and non-trivial:

The penetration curvature is:

$$K_{\text{penetration}} = \frac{1}{R_{\text{bone,eff}}} - 1$$

Inflamed bone is more vascular; $\text{CRP} > 100$ mg/L increases R_{bone} :

$$R_{\text{bone,eff}} = R_{\text{bone}} \cdot [1 + 0.006 \cdot \max(\text{CRP} - 100, 0)], \quad \text{capped at } 2 \times R_{\text{bone}}$$

This is why vancomycin attains adequate serum levels — $\text{AUC}/\text{MIC} > 400$ in many patients — and still fails in chronic osteomyelitis: the serum target and the bone target are not the same target.

Drug	Bone:Serum	$K_{\text{penetration}}$	Source
Vancomycin	0.10–0.30	2.33–9.00	Graziani et al., JAC 1988
Ceftaroline	0.20–0.40	1.50–4.00	Riccobene et al., AAC 2014
Linezolid	0.40–0.60	0.67–1.50	Rana et al., JAC 2002
Clindamycin	0.30–0.75	0.33–2.33	Feigin et al., Pediatr Infect 1995
Rifampin	0.20–0.50	1.00–4.00	Currier et al., JBJS 1979
Daptomycin	0.10–0.20	4.00–9.00	Traunmuller et al., AAC 2010

Table 1: Bone penetration ratios and resulting curvature contributions. $K_{\text{penetration}} = (1/R_{\text{bone}}) - 1$.

Layer 3: Biofilm Resistance Curvature

MRSA in established osteomyelitis forms biofilm — structured communities of phenotypically dormant bacteria embedded in extracellular matrix. Biofilm MICs are 100- to 1000-fold higher than planktonic MICs (Stewart, 2015). The minimum biofilm eradication concentration (MBEC) determines effective drug activity:

$$K_{\text{biofilm}} = \log_{10} \left(\frac{\text{MBEC}}{\text{MIC}} \right)$$

Drug	MIC ($\mu\text{g/mL}$)	MBEC ($\mu\text{g/mL}$)	K_{biofilm}	Source
Vancomycin	1.0	512	2.71	Parra-Ruiz et al., 2012
Clindamycin	0.25	64	2.41	LaPlante & Rybak, 2004
Ceftaroline	1.0	128	2.11	Barber et al., AAC 2015
Linezolid	2.0	256	2.11	Parra-Ruiz et al., 2012
Daptomycin	0.5	32	1.81	Stewart, 2015
Rifampin	0.008	0.5	1.80	Zimmerli et al., NEJM 2004

Table 2: MIC, MBEC, and biofilm curvature by drug. Rifampin has the lowest K_{biofilm} of any anti-staphylococcal agent — the geometric basis for its role in combination therapy for chronic AHO.

Biofilm probability is inferred from the clinical timeline:

$$p_{\text{biofilm}} = \begin{cases} 0.20 & \text{acute (< 2 weeks)} \\ 0.60 & \text{subacute (2 weeks–3 months)} \\ 0.95 & \text{chronic (> 3 months)} \end{cases} \quad K_{\text{biofilm,eff}} = p_{\text{biofilm}} \cdot K_{\text{biofilm}}$$

The MIC reported on an antibiogram is the planktonic MIC. The bacteria causing chronic osteomyelitis are in biofilm, with an effective MIC 100- to 1000-fold higher. The Keske Method computes this from published MBEC data and weights it by the probability that biofilm is present.

Layer 4: Multi-Reservoir Geometry

MRSA-AHO distributes across three anatomically distinct reservoirs with different antibiotic accessibility:

Reservoir	Location	Accessibility	Intervention
Soft tissue abscess	Periosteal / subperiosteal	High (vascularized)	Surgical drainage
Bone matrix biofilm	Cortical / cancellous surfaces	Moderate (penetration-limited)	Debridement
Intracellular SCVs	Inside osteocytes	Very low	Rifampin; no surgical access

Table 3: Three-reservoir model. Small colony variants (SCVs) inside osteocytes drive relapse in chronic AHO. They are not reached by surgery and are accessible only to drugs with intracellular penetration — principally rifampin.

The reservoir curvature term, per drug:

$$K_{\text{reservoir}} = (1 - D_{\text{drain}}) \cdot w_{\text{SAC}} + (1 - D_{\text{debride}}) \cdot K_{\text{penetration}} + 0.8 \cdot f_{\text{intra}} \cdot m_{\text{rif}}$$

where D_{drain} , D_{debride} are surgical clearance fractions (0–1), f_{intra} is the intracellular bacterial fraction, $w_{\text{SAC}} = 0.5$, and $m_{\text{rif}} = 0.4$ when rifampin is present in the regimen (reflecting its documented osteoblast penetration; Zimmerli 2004), else $m_{\text{rif}} = 1.0$.

Applied to a Clinical Scenario

Patient: 10-year-old, 32 kg. MRSA-AHO of the hip, femur, and knee. Sepsis and TSS on presentation; chronic recurrent course. Multiple antibiotics trialed without durable remission.

Input parameters:

$$\text{CRP} = 250 \text{ mg/L}, \quad p_{\text{biofilm}} = 0.95, \quad f_{\text{intra}} = 0.40, \quad D_{\text{drain}} = 0.90, \quad D_{\text{debride}} = 0.80$$

Computed C_{bone} for each drug — monotherapy. Every column is produced by the engine; $K_{\text{total}} = K_{\text{ADMET}} + K_{\text{pen}} + K_{\text{bio}} + K_{\text{res}}$ and $C_{\text{bone}} = \tau / K_{\text{total}}$.

Drug	τ	K_{ADMET}	K_{pen}	K_{bio}	K_{res}	K_{total}	C_{bone}
Linezolid	12	0.40	0.05	2.00	0.38	2.84	4.23
Daptomycin	24	0.60	2.51	1.72	0.87	5.70	4.21
Ceftaroline	12	0.67	0.75	2.00	0.52	3.95	3.04
Rifampin [†]	8	0.50	0.50	1.71	0.28	2.99	2.68
Clindamycin	8	0.50	0.00	2.29	0.37	3.16	2.53
Vancomycin	12	0.50	1.63	2.57	0.70	5.40	2.22

Table 4: Cure threshold: $C_{\text{bone}} \geq 5.0$. **No monotherapy achieves it** in chronic MRSA-AHO with confirmed biofilm — the best single agent reaches only 4.23. This matches the documented clinical course for chronic AHO, in which serologically adequate monotherapy fails. [†]Rifampin is never used as monotherapy (rpoB resistance emerges in ~80% of cases); its value is shown only as a combination component.

Combination therapy

No single agent reaches threshold, so the method scores a two-drug regimen. Each drug is first reduced to its own bone coherence $C = \tau/K_{\text{pathway}}$; the two are then combined by their *measured pharmacodynamic interaction* — not by a fixed multiplier that always helps. Three interaction types are representable, so an unfavorable pairing is not silently rewarded:

$$C_{\text{combo}} = \begin{cases} C_A + C_B & \text{additivity (Bliss independence)} \\ \max(C_A, C_B) + \delta & \text{synergy} \\ \min(C_A, C_B) \cdot (1 - \delta) & \text{antagonism} \end{cases}$$

The interaction type and its magnitude δ are *evidence inputs* (checkerboard / FICI / Bliss), supplied per isolate rather than assumed.

For **ceftaroline + rifampin** at this geometry — with rifampin’s intracellular reservoir term reduced ($m_{\text{rif}} = 0.4$) — the per-drug bone coherences are $C_{\text{ceftaroline}} = 3.20$ and $C_{\text{rifampin}} = 2.68$. Under Bliss independence:

$$C_{\text{bone}}(\text{cef} + \text{rif}) = 3.20 + 2.68 = \mathbf{5.87} \geq 5.$$

This is the first regimen in this patient’s geometry to cross the cure threshold — where vancomycin, the standard-of-care agent, scores 2.22. The result is a $2.6\times$ improvement in therapeutic coherence, and it comes from the two barriers no single agent overcomes: bone penetration and rifampin’s access to the intracellular reservoir.

The result is explicitly *contingent on the interaction*: additivity or synergy places the regimen at or above threshold, while strain-level antagonism — which has been reported for rifampin combinations in some isolates (Barber et al., AAC 2015) — would place it below. Determining the interaction for a given isolate, by checkerboard or FICI, is therefore the decisive measurement. For reference, an earlier “parallel-conductance” combination heuristic is retired in favor of this evidence-typed model, which cannot manufacture benefit from an inactive or antagonistic partner.

Clinical Implication: What the Math Tells Us

1. **Serum adequacy is not bone adequacy.** Vancomycin achieving $\text{AUC}/\text{MIC} > 400$ in serum is the wrong target when R_{bone} is low and $p_{\text{biofilm}} = 0.95$. That target was derived from bacteremia data, not osteomyelitis data. The Keske Method quantifies the bone-specific deficit prospectively rather than inferring it retrospectively from treatment failure.
2. **Biofilm is the dominant curvature term in chronic disease, and antibiograms do not measure it.** At $p_{\text{biofilm}} \geq 0.8$, no monotherapy crosses the cure threshold; a combination is what brings a regimen across it. Rifampin’s low K_{biofilm} and unique intracellular access make it the natural partner — consistent with its established role in refractory staphylococcal bone and implant infection (Zimmerli 2004). Whether a specific pairing reaches threshold depends on the measured drug-drug interaction, which the model takes as an input rather than assuming.
3. **Surgical intervention has a quantifiable pharmacodynamic effect.** Drainage and debridement reduce $K_{\text{reservoir}}$ and secondarily raise $R_{\text{bone,eff}}$ by reducing inflammatory mass.

The model computes the specific D_{drain} and D_{debride} values that, in combination with a given drug pair, bring C_{bone} to ≥ 5 — an operable, patient-specific surgical target rather than a generic recommendation.

4. **Intracellular reservoirs are the remission barrier.** At high f_{intra} , a regimen cannot hold threshold without a drug that has documented intracellular penetration; rifampin and linezolid are the principal anti-MRSA agents in that class. Sequential design — a cell-wall-active agent first to reduce biofilm mass, then a rifampin combination to address SCVs — is the direction the geometry supports.

What This Is — and What It Is Not

The Keske Method is an **investigational mathematical framework**. It is not an FDA-cleared diagnostic, dosing calculator, or prescribing tool. It is presented strictly for theoretical evaluation, retrospective study, and hypothesis generation by qualified clinical researchers. It does not replace susceptibility testing, infectious disease consultation, or clinical judgment. It takes no action autonomously and makes no therapeutic recommendation.

What it provides: a systematic, patient-specific, mathematically grounded answer to “*why didn’t that antibiotic work?*” — and a directional answer to “*what geometry do we need to reach threshold?*” Every figure in this brief is computed by the open engine and is reproducible from the stated inputs.

The live interactive tool (MIRADOR — Keske Edition) is publicly available at:

<https://mirador.vercel.app/#keske>

Repository and validation results: [doi:10.5281/zenodo.18511755](https://doi.org/10.5281/zenodo.18511755)

Medical and clinical patent applications (selected; Davis, B.R., sole inventor; serial numbers as filed):

1. Davis, B.R. *MIRADOR: Manifold-Informed Rational Architecture for Drug-Organism Response — Geometric Optimization of Therapeutic Coherence with Resistance Prediction via Escape Geodesics on Drug-Target Fiber Bundles*. U.S. non-provisional application (serial no. on file with the USPTO), 2026; claiming priority to a related provisional.
2. Davis, B.R. *TESSERA: Transfer-Enabled Segmented Surveillance of Emergent Resistance in Antimicrobials*. U.S. Provisional Patent Application No. 63/933,236.
3. Davis, B.R. *Genomic Evolution on Davis-Embedded Surfaces for Integrated Cancer Detection*. U.S. Provisional Patent Application No. 63/931,845.
4. Davis, B.R. *HERALD: High-Resolution Early Recognition of Antigenic Landscape Divergence*. U.S. Provisional Patent Application No. 63/919,595.
5. Davis, B.R. *Geometric Phase Transition Detection and Curvature-Based Resistance Emergence Threshold Identification*. U.S. Provisional Patent Application No. 63/926,936.

Proposed Retrospective Validation

The $C_{\text{bone}} \geq 5$ threshold and the curvature weights used throughout this model are derived from published pharmacokinetic, microbiological, and clinical data. They have not been prospectively validated in a clinical cohort. We propose the following validation design for researchers with access to institutional records:

Study design: Multi-center retrospective cohort of ≥ 100 cases of pediatric MRSA-AHO (age < 18 , culture-confirmed MRSA, minimum 12-month follow-up), classified as treatment success (remission at 12 months) or failure (relapse, recurrence, or chronic suppression).

Primary analysis: For each historical case, input the available clinical variables (admission CRP, infection duration, surgical interventions, antibiotic sequence) into the MIRADOR engine to compute C_{bone} for the regimen actually used. Test whether $C_{\text{bone}} < 5$ predicts clinical failure better than standard metrics (vancomycin AUC/MIC; MIC $\leq 1 \mu\text{g/mL}$ cutoff).

Hypothesis: Logistic regression will show the geometric impedance term K_{pathway} is a significant predictor of relapse independent of serum drug exposure, with AUROC > 0.80 against clinical outcome.

Data requirements: Admission CRP, weight, estimated GFR, antibiotic(s) used, surgical records, infection duration, and 12-month outcome — all available in standard EHR exports, requiring no additional testing or prospective enrollment. The engine accepts structured CSV input and is freely accessible at <https://mirador.vercel.app/#keske>.

Invitation

If this framework is of interest for clinical validation, collaborative study, or integration into pediatric AHO research protocols, I welcome the conversation.

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The Keske Method is named for Steven Keske, who was diagnosed with sepsis, toxic shock syndrome, and acute hematogenous osteomyelitis in April 2020 at age 10. He survived 10 days in the PICU, 4 surgeries, and 2 months on a PICC line. He has experienced recurrent infections for six years. He is still fighting. His mother Barbara shared his story publicly so that other families would know what to look for. That story became this framework.